nature portfolio

Peer Review File

Large Language Models Lack Essential Metacognition for Reliable Medical Reasoning

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
The paper 'Large Language Models' Deceptive Expertise: A Threat to Reliable Clinical Decision Support Systems' addressed a highly topical theme. The point that large language models (LLMs) have deceptive expertise is well known in many domains, not only medicine, in general life, in the media, and in comments and reports in the scientific literature. It is an important issue, and additional analysis to demonstrate the nature of this deceptive expertise is relevant. The paper is very well written throughout.

The paper explores large language models deceptive expertise. The authors added to the MedQA-USMLE benchmark with sets of wrong answers, and confidence scores and metacognitive tasks including missing answer recall (identifying when correct answers are absent), and unknown recall (recognising questions beyond their knowledge).

There is an earlier version of the paper here: https://arxiv.org/pdf/2406.02394

This included this question/answer set with a totally made up endocrine gland, the 'Glianorex' and made up medical imaging modalities 'Glianorex Imagery Sonography (GIS)'

I have no doubt that the paper is of interest and is worthy of publication in the peer reviewed literature after improvement in some areas.

Although the paper is of interest I do have some concerns about related to the depth to which its methodology, the extent to which it really establishes the problem described, and the actual degree of relevance to the current approaches by responsible actors in developing decision support systems.

Concerns/issues:

(1) The overall approach used in the manuscripts to take the base 'off the shelf' versions of LLMs without further fine tuning and to ask the question posed in the revised (study specific) question set does not explore in any meaningful way the sensitivity of the LLMs to the manner in which the question is posed. It could be that subtle variations in the exact manner that the model was prompted could result in substantially better, or even substantially worse performance, and a range of performance for each system depending on prompting.

(2) Related to the point above, it is unclear from the analysis, the extent to which a set of initial prompting in the manner of answering of medical information could resolve the issue addressed. For example, if the LLMs where pre-prompted to always be aware that the question could include deceptive, confused or incorrect information, and not to assume the posed medical constructs in the questions are relabel information - `I am thinking off the 'Glianorex' example.

(3) Although certain LLM-based decision support systems are appearing on the market at the moment (illegally or with questionable regulatory status), that seem to use relatively unfiltered LLMs Clinical Decision Support (CDS), it is likely that most CDS systems will use multiple layered approaches including retrieval augmented generation (RAG), and even LLM CDS models linked to non-LLM more traditional CDS approaches. Ideally the paper would at least discuss the implications of their findings on this approach, and the implications of multiple layer approaches on their findings. Again, thinking of the

Glianorex example, it would seem relatively straightforward to ask an LLM to pre-scan a question using RAG, and to highlight suspect medical information in the question to the clinician, in the areas where that information cannot be identified in the medical resources on which the RAG-process addresses.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)
In this study, the authors introduced a novel benchmark designed to evaluate the metacognitive abilities of twelve large language models (LLMs). The study involved augmenting multiple-choice questions with additional options to assess the models' capacity to recognize their areas of knowledge and limitations. The findings indicated that although the LLMs demonstrated high accuracy on multiple-choice questions, they failed to accurately identify their knowledge limitations. The models exhibited high confidence levels in their answers, even when no correct answer was present. The authors explored the implications of these results for clinical reasoning, emphasizing the potential risks of utilizing such models in clinical decision support without thorough evaluation.

Every health data is biased. Understanding the social patterning of the data generation process is key to unlocking the potential of LLMs. Please see:
Ghassemi M, Nsoesie EO. In medicine, how do we machine learn anything real? Patterns (N Y). 2022 Jan 14;3(1):100392. doi: 10.1016/j.patter.2021.100392. PMID: 35079713; PMCID: PMC8767288.

Metacognition models may be biased. A brief discussion of the shortcomings of traditional models of clinical reasoning is warranted if the authors are proposing to benchmark metacognition in how decisions are made in the healthcare domain. Please see:
Parsons AS, Wijesekera TP, Olson APJ, Torre D, Durning SJ, Daniel M. Beyond thinking fast and slow: Implications of a transtheoretical model of clinical reasoning and error on teaching, assessment, and research. Med Teach. 2024 Jun 5:1-12. doi: 10.1080/0142159X.2024.2359963. Epub ahead of print. PMID: 38835283.
Daniel, Michelle, Wilson, Eric, Seifert, Colleen, Durning, Steven J., Holmboe, Eric, Rencic, Jospeh J., Lang, Valerie and Torre, Dario. "Expanding boundaries: a transtheoretical model of clinical reasoning and diagnostic error" Diagnosis, vol. 7, no. 3, 2020, pp. 333-335. https://doi.org/10.1515/dx-2019-0102

Expansion of the discussion on metacognition in LLMs and in clinical practice: The discussion section, while insightful, could be strengthened by a more thorough exploration of the meta-cognitive capabilities of language models in clinical practice. Specifically, the authors could consider expanding the discussion to include a more complex perspective on whether metacognition should be expected from LLMs. If so, it would be valuable to explore how this concept is defined from the perspective of physicians and how it could be effectively emphasized in the functioning of LLMs.
Examples of relevant references on metacognition as reflected in clinical reasoning:
An important education paper focused on promoting strong diagnostic reasoning skills among physicians - Bowen JL. Educational strategies to promote clinical diagnostic reasoning. N Engl J Med. 2006 Nov 23;355(21):2217-25. doi: 10.1056/NEJMra054782. PMID: 17124019.
An article on key features in assessing clinical decisions - Bordage G, Page G. Key features to assess clinical decisions. Med Teach. 2018 Nov;40(11):1195-1196. doi: 10.1080/0142159X.2018.1512746. Epub 2018 Nov 13. PMID: 30422035.
A comprehensive review on diagnostic reasoning and medical problem solving - Elstein AS. Thinking about diagnostic thinking: a 30-year perspective. Adv Health Sci Educ Theory Pract. 2009 Sep;14 Suppl 1:7-18. doi: 10.1007/s10459-009-9184-0. Epub 2009 Aug 11. PMID: 19669916.
The interplay between uncertainty and confidence in medical decision-making - Zwaan L, Hautz WE. Bridging the gap between uncertainty, confidence and diagnostic accuracy: calibration is key. BMJ Qual Saf. 2019 May;28(5):352-355. doi: 10.1136/bmjqs-2018-009078. Epub 2019 Feb 6. PMID: 30728186.
On reflective practice in medicine - Mamede S, Schmidt HG. The structure of reflective practice in medicine. Medical Education [Internet]. 2004 Dec;38(12):1302–8. Available from: https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1365-2929.2004.01917.x PMID15566542
Additional article on cognitive biases - Croskerry P. The importance of cognitive errors in diagnosis and strategies to minimize them. Acad Med. 2003 Aug;78(8):775-80. doi: 10.1097/00001888-200308000-00003. PMID: 12915363.

Lack of recent references on meta-cognition in LLMs: The manuscript would benefit from incorporating recent and relevant references on metacognition in language models and its application in clinical decision-making. This is essential for providing a comprehensive and current understanding of the field. The lack of such references limits the analysis's depth and risks overlooking key advancements and discussions that could enhance the study's conclusions. Recent papers t
While GPT-4V demonstrated high accuracy, its rationales often contained errors, especially in image comprehension, indicating that correct answers may not always stem from sound reasoning. The study emphasizes the need for comprehensive evaluations beyond accuracy to ensure effective integration of AI models into clinical practices. Jin Q, Chen F, Zhou Y, Xu Z, Cheung JM, Chen R, Summers RM, Rousseau JF, Ni P, Landsman MJ, Baxter SL, Al'Aref SJ, Li Y, Chen A, Brejt JA, Chiang MF, Peng Y, Lu Z. Hidden flaws behind expert-level accuracy of multimodal GPT-4 vision in medicine. NPJ Digit Med. 2024 Jul 23;7(1):190. doi: 10.1038/s41746-024-01185-7. PMID: 39043988; PMCID: PMC11266508.

Enhancement of the methods section with additional regression analysis: The methods section could be significantly improved by incorporating regression analysis to examine the relationship between model accuracy and reasoning ability. This statistical approach would allow for a more nuanced understanding of whether more accurate models also exhibit superior meta-cognitive capabilities, thus providing a deeper insight into the potential of these models for clinical reasoning tasks.

Broadening the assessment of reliability and reproducibility: The current approach to assessing the reliability and reproducibility of the models, which focuses primarily on temperature settings, is somewhat limited. To provide a more robust evaluation, it would be beneficial to include additional techniques such as top-p and top-k sampling. These methods are recognized in the field and by incorporating them in the study process, the authors could offer a more comprehensive assessment of the models' performance and reliability.

(Remarks on code availability)
The manuscript does not present a model that needs to be assessed for reproducibility, but rather a framework to assess the "reasoning" behind an LLM output.

Reviewer #3

(Remarks to the Author)
In the paper presented, the authors motivate the use of LLMs for clinical decision support systems. In particular, to explore metacognition, the authors have developed a benchmark, MetaMedQA - the link to the code unfortunately returns an error 404. The paper itself is interesting and well written, the main finding lies in a discrepancy between the perceived and actual capabilities of LLMs in medical reasoning, thus revealing their deceptive expertise, which is good. This overconfidence actually poses significant risks in clinical settings where recognizing uncertainty is of paramount importance. The inability of these models to recognize knowledge gaps calls into question their suitability for integration into clinical decision support systems, which is undoubtedly true.
The work itself shows that the AI readiness for clinical use, especially in decision support, is greatly overestimated by the current evaluation methods. The submitted work addresses an important topic, is interesting, relevant, fits well into the chosen journal and is also well written. The reviewer trusts the authors regarding the software used. For all the reasons mentioned, this reviewer is cautiously positive and recommends possible acceptance, but suggests a few minor improvements to further increase the benefit for the interested reader.
1) This work is a good and representative example of the use of hyper AI in medicine in general, and two important things should be added for the interested reader (preferably in the introduction):
AI always has (positive) and (negative) effects:
a) The potential (positive) effects of AI in healthcare, which include diagnosis and treatment, the provision of healthcare services and public health initiatives, cannot be overestimated, and despite all the criticism, a significant health benefit should also be achieved. Predictive models that enable the early and precise detection of diseases offer enormous possibilities, but as we know, they are largely not used in practice. Thus, AI has great potential but is still underutilized in clinical medicine; this should be mentioned and there is a very recent (brand new - this reviewer has just discovered and read this work and it fits perfectly here) paper that has examined exactly this topic and how the healthcare system can deal with the underuse of AI, otherwise overregulation threatens to forego the potential benefits of AI, see this highly topical paper:
Pagallo, U. et al. 2024. The underuse of AI in the health sector: Opportunity costs, success stories, risks and recommendations. Health and Technology, 14, 1--14, doi:10.1007/s12553-023-00806-7.
b) At the same time, however, attention should be drawn to the (negative) effects, and that is the real reason why we need traceability, interpretability and explainability, namely to ensure in the event of damage why something was decided in a certain way, so it's about trust - which is unfortunately not mentioned here at all, so it should at least be mentioned and the following highly cited and very well-known reference, which describes very well that the European Commission already has new legislation for the use of “high-risk artificial intelligence” and that the existing European framework of fundamental rights already provides some clear guidelines for the use of medical AI, which will definitely become important for its use in medicine in the future, see: Stoeger, K. et al. 2021. Medical Artificial Intelligence: The European Legal Perspective. Communications of the ACM, 64, (11), 34--36, doi:10.1145/3458652

(Remarks on code availability)
This reviewer tried to check the code, however, with
https://github.com/maximegmd/metamedqa-benchmark
one gets a 404 error ...

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

My 2nd review comments are short.

The methodology and results from the prompt engineering analysis are really interesting and in my view add to the paper. I do not request any changes at this stage.

The additional analysis (regression) asked for by one of the other reviewers adds depth to the paper. I do not request any changes at this stage.

There are good descriptive additions setting the work in context with recent literature.

(Remarks on code availability)
NA

Reviewer #2

(Remarks to the Author)
Thank you for the opportunity to review the revised manuscript and the rebuttal to the reviewers' comments. We are satisfied with the revisions and explanations.

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)
In the opinon of this reviewer, the authors have adequately addressed all the reviewers' comments, consequently this reviewer would now argue to accept this paper.

(Remarks on code availability)
all good

Response to comments

Dear reviewers, thank you for your detailed comments and suggestions which greatly improved the quality of our manuscript. Please find below, a detailed response to each comment followed by a list of changes made to implement the suggestions.

We first address the availability of the source code, we were under the impression that the code was made available to the reviewers through the Code Ocean capsule submission along with the paper at the following address https://codeocean.com/capsule/0254556/tree and therefore kept the GitHub repository private until the review process completed. We have made the GitHub public and added the updated code for the new experiment. (https://github.com/maximegmd/MetaMedQA-benchmark).

In addition, OpenAI released O1-preview during this revision time, sadly we still do not have API access to this model and therefore were not able to evaluate this new model.

Finally, we reformatted the manuscript to adhere to the formatting instructions of the journal.

Reviewer #1

Dear reviewer #1, thank you for your insightful comments which are mainly focused on additional engineering to alleviate the issues identified in this work. We respond to each of your concern below but want to address the main idea first.

We understand the importance of engineering solutions and agree that models by themselves should not be deployed or used in healthcare settings without additional layers of security. Sadly, many healthcare organizations make standalone models available through solutions such as Microsoft 365 Copilot.

In this work, we mainly focus on the base LLM models to put into perspective issues that have not been identified or measured in the context of medicine. The scope of this study is to demonstrate these limitations and propose a methodology to measure such limitations. We are of the opinion that exploring additional engineering methods in our fundamental analysis would introduce a bias and would be difficult to justify due to performance differences that could be explained by the difference of sensitivity to specific prompts.

But we agree that addressing potential solutions is important and have performed additional experiments with GPT-4o which we evaluated iteratively with different prompts with the goal of improving the model's performance on the "missing answer recall" and "unknown recall" metrics.

(1) The overall approach used in the manuscripts to take the base 'off the shelf' versions of LLMs without further fine tuning and to ask the question posed in the revised (study specific) question set does not explore in any meaningful way the sensitivity of the LLMs to the manner in which the question is posed. It could be that subtle variations in the exact manner that the model was prompted could result in substantially better, or even substantially worse performance, and a range of performance for each system depending on prompting.

Thank you for your remark on prompting dependence. We agree that pre-prompting and additional techniques such as MedPrompt[1] or few short learning[2] improve the performance of models on benchmarks. But the goal of our study is to assess the base capabilities without additional engineering. Similar work conducted on bias assessment also evaluated the models without additional prompt engineering or finetuning[3].

It is standard practice to use 0-shot for answering multiple choice questions without prior prompt engineering to ensure a fair comparison between models that may be sensitive to specific prompts. This methodology is the default implementation in LM evaluation harness[4]

[1] Can Generalist Foundation Models Outcompete Special-Purpose Tuning? Case Study in Medicine. Nori et al. arXiv:2311.16452

[2] Language Models are Few-Shot Learners. Brown et al. Advances in Neural Information Processing Systems 33 (NeurIPS 2020)

[3] Assessing the potential of GPT-4 to perpetuate racial and gender biases in health care: a model evaluation study. Zack, Travis et al. The Lancet Digital Health, Volume 6, Issue 1, e12 - e22

[4] Lessons from the Trenches on Reproducible Evaluation of Language Models. Birderman et al. arXiv:2405.14782

and the proposed methodology by the authors of the MedQA[5] benchmark which we based MetaMedQA on.

Ideally, we would have sent the questions directly without any prompt but, we were forced to use a basic prompt engineering technique to ensure models such as GPT-4 that do not output logits - and therefore cannot be constrained - generate their response in the expected format. We used the same prompt for every model even those that produced logits:

"You must only respond in the following format 'Answer: A/B/C/D/E/F' and then 'Confidence: 1/2/3/4/5' on a scale from 1 being not confident at all to 5 being absolutely sure."

See response to (2) for details on the additional experiment conducted to assess how prompt engineering affects the performance of models on our benchmark.

(2) Related to the point above, it is unclear from the analysis, the extent to which a set of initial prompting in the manner of answering of medical information could resolve the issue addressed. For example, if the LLMs where pre-prompted to always be aware that the question could include deceptive, confused or incorrect information, and not to assume the posed medical constructs in the questions are relabel information - `I am thinking off the 'Glianorex' example.

We performed an additional experiment using GPT-4o and a variety of system prompts to assess their impact. We found that warning the model about potential tricky questions was not enough to generate a significant improvement on scores. But, explicitly telling the model that some questions may be impossible to answer was able to improve significantly the performance on "unknown recall" reaching 51.8% for the best prompt. As for "missing answer recall", while we did notice an improvement when explicitly telling the model that some questions may have missing answers, the improvement was not statistically significant. Interestingly, providing the model with the methodology used to create the benchmark had a relatively small effect compared to the explicit prompts.

This experiment demonstrates that prompt engineering can reduce the deficiencies uncovered by our study but need to be explicit, making the use of prompt engineering impractical in real case scenarios where many deficiencies need to be compensated, some of which may not be known at the time of design of the system. However, such improvement suggests that current LLMs could be leveraged to generate synthetic datasets to improve the alignment of LLMs on metacognitive tasks.

(3) Although certain LLM-based decision support systems are appearing on the market at the moment (illegally or with questionable regulatory status), that seem to use relatively unfiltered LLMs Clinical Decision Support (CDS), it is likely that most CDS systems will use multiple layered approaches including retrieval augmented generation (RAG), and even LLM CDS models linked to non-LLM more traditional CDS approaches. Ideally the paper would at least

[5] What Disease Does This Patient Have? A Large-Scale Open Domain Question Answering Dataset from Medical Exams. Jin et al. Applied Sciences. 2021; 11(14):6421.

discuss the implications of their findings on this approach, and the implications of multiple layer approaches on their findings. Again, thinking of the Glianorex example, it would seem relatively straightforward to ask an LLM to pre-scan a question using RAG, and to highlight suspect medical information in the question to the clinician, in the areas where that information cannot be identified in the medical resources on which the RAG-process addresses.

Thank you for suggesting addressing RAG, we agree with reviewer #1 that these systems can improve performance and could be used to detect suspicious information. We believe RAG could be seen as an external database models can interact with and therefore be included in the transtheoretical model of cognition referenced by reviewer #2. We expanded our discussion to address the effect of external tools to reduce the shortcomings of LLMs identified in our study.

Actions

Based on reviewer #1's feedback and concerns we performed an additional experiment using GPT-4o to assess the impact of prompt engineering on the performance on the metacognitive tasks evaluated in our newly introduced benchmark.

- Added the methodology of this new experiment.
- Added a table containing all system prompts evaluated.
- Added the results including a table reporting all metrics for each prompt along with statistical significance.
- Discuss the relevance of prompt engineering to alleviate these shortcomings.
- Discuss the use of RAG as a component of cognition as described in the transtheoretical model of cognition referenced by reviewer #2.

Reviewer #2

Dear reviewer #2, thank you very much for the extensive literature recommendations, we greatly appreciate your suggestions and believe that the improvements add depth to our analysis.

Every health data is biased. Understanding the social patterning of the data generation process is key to unlocking the potential of LLMs. Please see:

Thank you for your article recommendation which provided additional insight, we expand our discussion suggesting that the training process and data used which favour positive responses could explain the inability of models to express negatives.

Metacognition models may be biased. A brief discussion of the shortcomings of traditional models of clinical reasoning is warranted if the authors are proposing to benchmark metacognition in how decisions are made in the healthcare domain. Please see:

The metacognitive models used in our study are indeed relatively simple and addressing the shortcomings of this model is warranted. Thank you for the literature recommendations.

We added additional content in the limitations to address our focus on system 1 for the dual processing theory. We also address the limitations of the dual processing theory considering more complex models such as the transtheoretical model of clinical reasoning.

Expansion of the discussion on metacognition in LLMs and in clinical practice: The discussion section, while insightful, could be strengthened by a more thorough exploration of the meta-cognitive capabilities of language models in clinical practice. Specifically, the authors could consider expanding the discussion to include a more complex perspective on whether metacognition should be expected from LLMs. If so, it would be valuable to explore how this concept is defined from the perspective of physicians and how it could be effectively emphasized in the functioning of LLMs.

We agree that the discussion would benefit from additional exploration of what the expectations of LLMs in clinical practice is. We included the suggested literature and improved the discussion to justify why metacognition should be expected of LLMs to have a place in clinical practice.

Lack of recent references on meta-cognition in LLMs: The manuscript would benefit from incorporating recent and relevant references on metacognition in language models and its application in clinical decision-making. This is essential for providing a comprehensive and current understanding of the field. The lack of such references limits the analysis's depth and risks overlooking key advancements and discussions that could enhance the study's conclusions.

We agree with reviewer #2 that addressing current understanding of metacognition of LLMs would enhance our conclusions. We have added the suggested literature and improved the discussion and limitations sections of our work to address the current understanding of LLM cognition and how it relates to human processes of cognition.

Enhancement of the methods section with additional regression analysis: The methods section could be significantly improved by incorporating regression analysis to examine the relationship between model accuracy and reasoning ability. This statistical approach would allow for a more nuanced understanding of whether more accurate models also exhibit superior meta-cognitive capabilities, thus providing a deeper insight into the potential of these models for clinical reasoning tasks.

Thank you for this suggestion. We've implemented regression and correlation analyses for missing answer recall and unknown recall, adding these to the methods section and including a new figure.

For missing answer recall, the regression provides valuable insights, showing accuracy as a strong predictor of this meta-cognitive capability. We discuss these new results and their implications.

For unknown recall, we couldn't meaningfully interpret the regression as 9 out of 12 models scored 0 on this metric. We've noted this limitation in the results.

Broadening the assessment of reliability and reproducibility: The current approach to assessing the reliability and reproducibility of the models, which focuses primarily on temperature settings, is somewhat limited. To provide a more robust evaluation, it would be beneficial to include additional techniques such as top-p and top-k sampling. These methods are recognized in the field and by incorporating them in the study process, the authors could offer a more comprehensive assessment of the models' performance and reliability.

Thank you for suggesting the use of Nucleus and top-k sampling in addition to temperature. In this study, we used constrained decoding to ensure the models' outputs are within the expected options. This implies that the sampler does not consider the tokens that are not within the expected grammar. In our case the grammar can either be [‘A’, ‘B’, ‘C’, ‘D’, ‘E’, ‘F’] for answer choices or [‘1’, ‘2’, ‘3’, ‘4’, ‘5’] for confidence score. The sampler will then use the temperature to sample a token from the subset of tokens in the grammar. We opted for a temperature of 0 to ensure deterministic results through the consistent sampling of the token with the highest probability.

We have clarified that we used constrained decoding in the methods.

Reviewer #3

Dear reviewer #3, thank you for your interest in our work and suggestions to improve our introduction and background. We would like to apologize for the miscommunication of the source code which we believed would be transferred to reviewers through the Code Ocean capsule submitted with the manuscript.

a) The potential (positive) effects of AI in healthcare, which include diagnosis and treatment, the provision of healthcare services and public health initiatives, cannot be overestimated, and despite all the criticism, a significant health benefit should also be achieved. Predictive models that enable the early and precise detection of diseases offer enormous possibilities, but as we know, they are largely not used in practice. Thus, AI has great potential but is still underutilized in clinical medicine; this should be mentioned and there is a very recent (brand new - this reviewer has just discovered and read this work and it fits perfectly here) paper that has examined exactly this topic and how the healthcare system can deal with the underuse of AI, otherwise overregulation threatens to forego the potential benefits of AI, see this highly topical paper:

Pagallo, U. et al. 2024. The underuse of AI in the health sector: Opportunity costs, success stories, risks and recommendations. Health and Technology, 14, 1--14, doi:10.1007/s12553-023-00806-7.

Thank you for your feedback and suggestions to enhance our introduction and background. We appreciate your insights and have carefully considered your recommendations. We have made several revisions to address your comments and strengthen our manuscript.

We agree that highlighting the positive potential of AI in healthcare is crucial. As you suggested, we have expanded our introduction to emphasize the significant health benefits that AI can offer, particularly considering the healthcare worker shortage and administrative burden challenges faced by clinicians. We have also incorporated the concept of AI's current underutilization in clinical medicine, as you pointed out.

b) At the same time, however, attention should be drawn to the (negative) effects, and that is the real reason why we need traceability, interpretability and explainability, namely to ensure in the event of damage why something was decided in a certain way, so it's about trust - which is unfortunately not mentioned here at all, so it should at least be mentioned and the following highly cited and very well-known reference, which describes very well that the European Commission already has new legislation for the use of "high-risk artificial intelligence" and that the existing European framework of fundamental rights already provides some clear guidelines for the use of medical AI, which will definitely become important for its use in medicine in the future, see: Stoeger, K. et al. 2021. Medical Artificial Intelligence: The European Legal Perspective. Communications of the ACM, 64, (11), 34--36, doi:10.1145/3458652

We appreciate your emphasis on the importance of trust in AI systems and the need for traceability, interpretability, and explainability. In response to your feedback, we have expanded our discussion to include these critical aspects. We have also incorporated the reference to Stoeger et al. (2021) to highlight the European Commission's approach to regulating high-risk AI in healthcare and the existing framework of fundamental rights that guide medical AI use.